0.1 Isometry

Definition 0.1.1. Let (X, d_X) and (Y, d_Y) be Metric Spaces. A map

$$f : X \to Y \ \text{ s.t } \ \forall x_1, x_2 \in X, \ d_X(x_1, x_2) = d_Y(f(x_1), f(x_2))$$

is called an *Isometry*.

Theorem 0.1.2. Every isometry is an embedding.

Proof. Follows the steps.
1) f **is continuous:**
Let $x \in X$ be fixed. Given $\varepsilon > 0$, let $\delta = \varepsilon$. Then,

$$d_X(x,t) < \delta \implies d_Y(f(x), f(t)) = d_X(x,t) < \delta = \varepsilon,$$

so f is continuous.
2) f **is one-to-one**

$$f(x_1) = f(x_2) \iff d_Y(f(x_1), f(x_2)) = 0 \iff d_X(x_1, x_2) = 0 \iff x_1 = x_2.$$

3) f **is an open map from** X **to** $f[X]$**:**
Let $B_r(x) \subseteq X$ be an open ball in X. We show $f[B_r(x)]$ is open in the subspace $f[X]$.
Given $y \in f[B_r(x)]$, we have $y = f(t)$ for some $t \in B_r(x)$. Put $\varepsilon = r - d_X(x,t) > 0$.
Then, we claim that:

$$y = f(t) \in B_\varepsilon(f(t)) \cap f[X] \overset{(*)}{\subseteq} f[B_r(x)]$$

The inclusion $(*)$ is shown as follows:
let $u \in B_\varepsilon(f(t)) \cap f[X]$. Since $u \in f[X]$, $u = f(s)$ for some $s \in X$, and $d_Y(f(t), f(s)) < \varepsilon$.
By the definition of isometry, $d_X(t,s) = d_Y(f(t), f(s)) < \varepsilon$. Thus,

$$d_X(x,s) \le d_X(x,t) + d_X(t,s) < d_X(x,t) + \varepsilon = d_X(x,t) + (r - d_X(x,t)) = r.$$

This implies $s \in B_r(x)$, so $u = f(s) \in f[B_r(x)]$. This proves $(*)$.
Since every point $y \in f[B_r(x)]$ has a neighborhood in $f[X]$ contained in $f[B_r(x)]$, f is an open map from X to $f[X]$.
Therefore, f is an embedding. To simple, observe that:

$$f[B_r(x)] = B_r(f(x)) \cap f[X]$$

□

Theorem 0.1.3. Let (X, d) be a Compact Metric Space. Then, an isometry $f : X \to X$ is a Homeomorphism.

Proof. Since every isometry is embedding, enough to show that f is onto. Suppose that f is not onto map. That is,

$$X \setminus f[X]$$

is not empty set. Let $x_0 \in X \setminus f[X]$ be given. Define inductively:

$$x_{n+1} \overset{\text{def}}{=} f(x_n), \ n \ge 0$$

Since X is a compact space and f is continuous, $f[X]$ is compact subset of X, preserved by continuous.
And the Metric Space is Hausdorff, so $f[X]$ is closed, being compact set in the Hausdorff Space.
Therefore, there exists $r > 0$ such that $x_0 \in B_r(x_0) \subseteq X \setminus f[X]$.
Since $x_m \in f[X]$ for all $m \ge 1$, $x_m \notin B_r(x_0)$, or $d(x_m, x_0) \ge r$. Now, by the construction, for any $m > n \ge 0$,

$$d(x_m, x_n) = d(f(x_{m-1}), f(x_{n-1})) = d(x_{m-1}, x_{n-1}) = \cdots = d(x_{m-n}, x_0) \ge r$$

This implies $\{x_n\}$ cannot have a convergent subsequence, this is contradiction to X is a compact metric space. □

Comment: Infintie subset of compact space has at least one limit point. Therefore, there is a convergent subsequence.